

An overview of aeronautical telecommunications in Europe and world-wide

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Since the days of Alexander Graham Bell, telecommunications systems have grown in complexity and are reaching the point where calls can be made and received almost anywhere. One of the last peaceful havens away from the telephone has been the airline seat. In recent years this has been under attack from the engineers and marketeers responsible for the development and deployment of aeronautical telecommunications systems.

This paper provides an overview of two key aeronautical telecommunications technologies in which BT holds an interest — Skyphone and Terrestrial Flight Telecommunications System (TFTS). The former predates the latter and uses geostationary satellite technology to provide global communications (primarily) to long-haul aircraft, while the latter employs ground-based cellular technology to provide a lower cost service in limited geographical areas (primarily) to short and medium haul aircraft. (The Skyphone system offers calls at \$10 per minute whilst TFTS is currently priced at around half this rate.)

The paper also briefly covers the commercial usage of the two systems to date including deployment on aircraft and services utilised by passengers. The paper ends by looking to the future of the two systems by examining future service developments and the use to which these services may be put.

1. The Skyphone system

1.1 Introduction

Skyphone is a consortia of BT, TeleNor and SingTel which has been providing global satellite communications services to the aviation community since 1990. The world's first scheduled passenger telephone service was launched on Singapore Airlines in 1991. Since then, Skyphone has been at the forefront of new developments and was the first service provider to launch aeronautical satellite facsimile and X.25 services.

Skyphone's role is both as a service provider, selling air-time, and as a network operator providing the ground earth stations (GESs) required to interconnect with the terrestrial network. The Skyphone partners currently operate a total of seven GESs located in England, Norway and Singapore.

The Skyphone network conforms to the standard for aeronautical satellite services defined by the International Mobile Satellite Organisation (Inmarsat) [1]. Inmarsat is also responsible for providing and operating the geostationary satellites used by the Skyphone service. These satellites are located at 35 658 km above the equator and provide coverage up to latitudes 86 ° north and 86 ° south.

Inmarsat is a London-based global partnership of 79 telecommunications companies of which BT is the second major shareholder. Established in 1979, to provide communications services to the maritime community its charter was extended to include the provision of services to the aeronautical community in 1989.

For global coverage from a geostationary orbit a minimum of three satellites are required. The Inmarsat system is currently using four primary satellites which are located over the Pacific, Indian and Atlantic (two) oceans. Figure 1 illustrates the coverage for each of the satellites.

1.2 Services and facilities

The Inmarsat aeronautical satellite (AES) system has been designed to support both passenger, airline operational and air traffic communications services. An early requirement was the provision of a datalink for relaying of operational and safety messages to provide information on the aircraft's status. This has been developed into a full packet data service which is compatible with existing X.25 packet data networks.

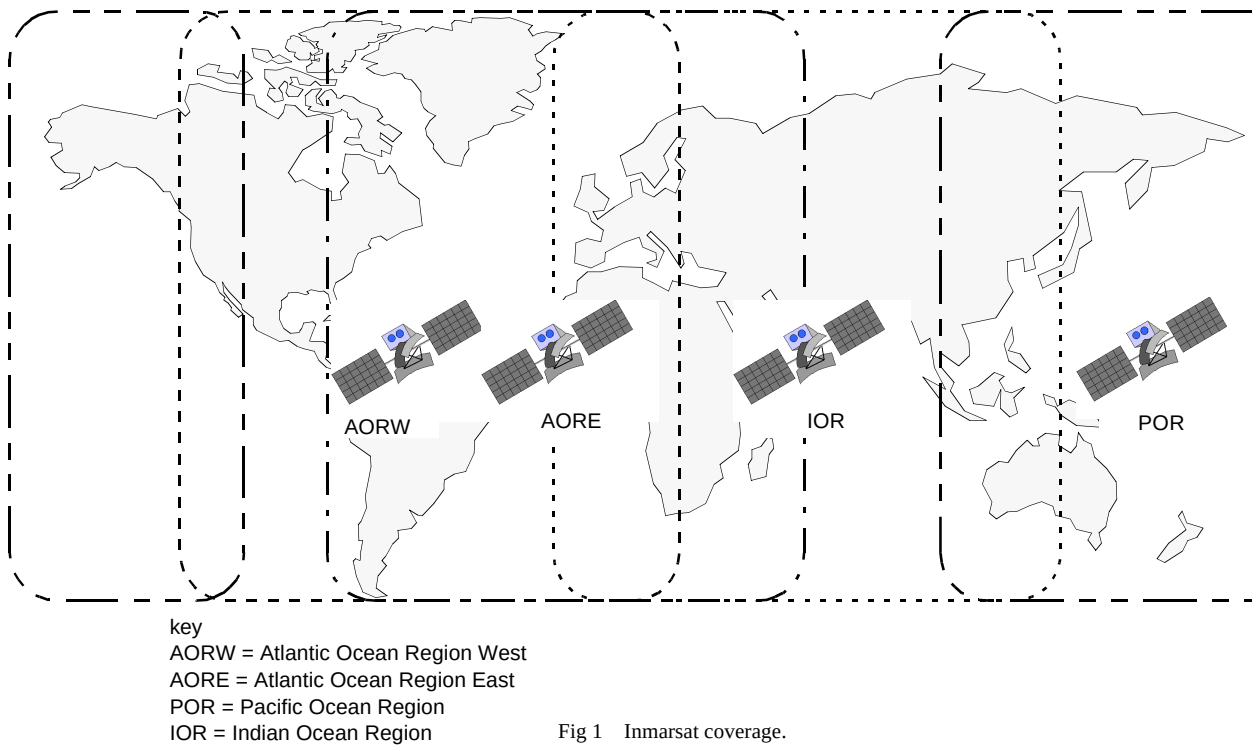


Fig 1 Inmarsat coverage.

Recently, airlines have become aware of the revenue opportunities in using the datalink for providing new passenger services including in-flight shopping and gaming. To expedite this development and to simplify the implementation of services across diverse computer platforms, BT has developed the AirLine Information Service (ALIS) system. ALIS provides a range of services including shopping, latest news and information, hotel, car, theatre and airline reservations services.

ALIS provides application software on board the aircraft which supports the passenger interface and processes all transactions. Transactions are forwarded to third-party suppliers via the ALIS central processor which connects to the Skyphone GESs via X.25 links. The system is currently in operation on Singapore Airlines and will be introduced on BA and Lufthansa during 1996.

In addition to packet data services the system supports voice, facsimile and circuit-mode data services. Facsimile services are offered at rates up to 4.8 kbit/s using V.27ter and circuit-mode data up to 2.4 kbit/s using V.22bis.

For all services, the system employs prioritisation of message which ensures that safety and operational calls take precedence over passenger calls. In normal operation this is imperceptible to the customer as a single aircraft installation can support five voice calls and multiple packet mode data calls.

1.3 Principal system features

The Inmarsat aeronautical system may be considered as a forerunner of today's terrestrial digital mobile systems such as GSM. The system exploits digital modulation and voice coding technology to make optimum use of the limited bandwidth and power available on satellite links with aircraft.

The present voice codec was selected by Inmarsat and the Airlines Electronic Engineering Committee (AEEC) in 1989 following the evaluation of 19 different designs. The requirement was that the codec should provide voice quality similar to that experienced by users of existing analogue mobile systems but using a bit rate of 9.6 kbit/s and with a channel bit error rate of 1 in 103. The design selected was developed at BT Laboratories (BTL) and uses the linear predictive coding (LPC) technique which exploits knowledge of the physiological and acoustic mechanisms of speech production [2]

In 1992, Inmarsat awarded BT a contract to develop an enhanced codec which could support facsimile and speech band data. BT's solution uses fast Fourier transform (FFT) spectral analysis techniques to distinguish between speech band data and speech. Depending on the outcome of the analysis the incoming data is routed either to the LPC codec, V.21/V.27ter modems for facsimile or V.22bis modem for circuit-mode data [3]. The solution thus provides a transparent mechanism for the support of circuit-

mode data services and requires no hardware upgrades in either the avionics or ground systems.

Another similarity with today's digital system is the need for the mobile terminals to register with the network before being allowed to use the system. This procedure is referred to as 'Log-on' and ensures that only authorised mobiles can access the system. The mechanism also means that it is technically possible to establish ground-to-air calls, although in practice most airlines do not permit this.

To satisfy an air traffic services requirement that the system should not have a single point failure, each GES operates in a stand-alone mode. To support this mode of operation, each GES is assigned a set of frequencies for its own use. The initial system was designed on the assumption that there would be only a small number of GESs; now that Inmarsat is facing a growth in the number of GESs, measures are being planned to avoid the potential inefficiencies in use of spectrum and power.

To improve spectrum efficiency Inmarsat have defined a network channel management system (NCMS). The NCMS role is to manage a central pool of voice channel frequencies which can be assigned to any GES in the system on demand. To conserve the amount of satellite power used, Inmarsat have defined that only a sub-set of GESs, referred to as Type I GES, will be required to provide the full range of aeronautical services. All other GESs, referred to as Type II GES, will provide the core voice and data services. These changes are anticipated to reduce the loading on the Inmarsat III satellites by as much as 10%.

The stand-alone mode of operation imposes no service constraints since calls can be placed and received through any GES in the system once an aircraft has logged on.

The continuity of the satellite services offered to a particular AES depends upon several elements, such as its current geographical position, the antenna performance and the aircraft trajectory. As a result an AES will typically log-on and log-off several times during a flight. The system does not support handover of voice and data calls, although the AES will delay handover if a call is in place.

1.4 System architecture

Figure 2 shows the overall architecture of the Skyphone network. Voice and data calls are transmitted from aircraft via Inmarsat satellites to the Skyphone GESs and then switched through international telecommunications networks to anywhere in the world.

Two types of avionics systems are currently supported and are referred to as Aero-H and Aero-L. Aero-H denotes the use of a high gain 12 dBi steerable antenna and has the following features:

- multiple, simultaneous voice calls,
- facsimile — Group 3 at 4.8 kbit/s,
- circuit-mode data services — V.22bis at 2.4 kbit/s,
- real time, two-way packet data services at 10.5 kbit/s.

Aero-L denotes the use of a low gain 0 dBi omnidirectional antenna and has the following features:

- real time, two-way packet data services at 600 bit/s.

A typical aircraft installation may include both Aero-H and Aero-L systems with the latter being used in the event of a failure of the high-gain system.

1.5 Radio interface structure

The aeronautical satellite system radio interface consists of TDM/TDMA, slotted Aloha and SCPC channels operating at L and C band. L band is used for air-to-satellite transmissions and C band is used for ground-to-satellite transmissions.

The following are the world-wide L band frequency allocations for aeronautical satellite use:

1544 to 1554 MHz	ground-to-air use
1655.5 to 1665.5 MHz	air-to-ground use

The 2×10 MHz spectrum is capable of supporting in excess of 550 voice channels each of 17.5 kHz, but this is limited by the available satellite power. The current second generation of Inmarsat satellites have available a total L band EIRP of 39 dBW, corresponding to just 90 simultaneous calls in the worst-case scenario. In practice, the number of calls that can be supported is much greater since the system uses sophisticated power control management which can reduce the L band channel EIRP required by as much as 10 dB.

Four types of RF channels are defined between the aircraft and the ground stations; these are referred to as P, R, T and C channels and are summarised in Table 1 and in Fig 3.

- P channels

The P channel is fundamental to the operation of the system since an aircraft is unable to contact a GES until it has received the latest version of the system broadcast table. This table contains information on all GESs in the system, the status and location of satellites and access frequencies.

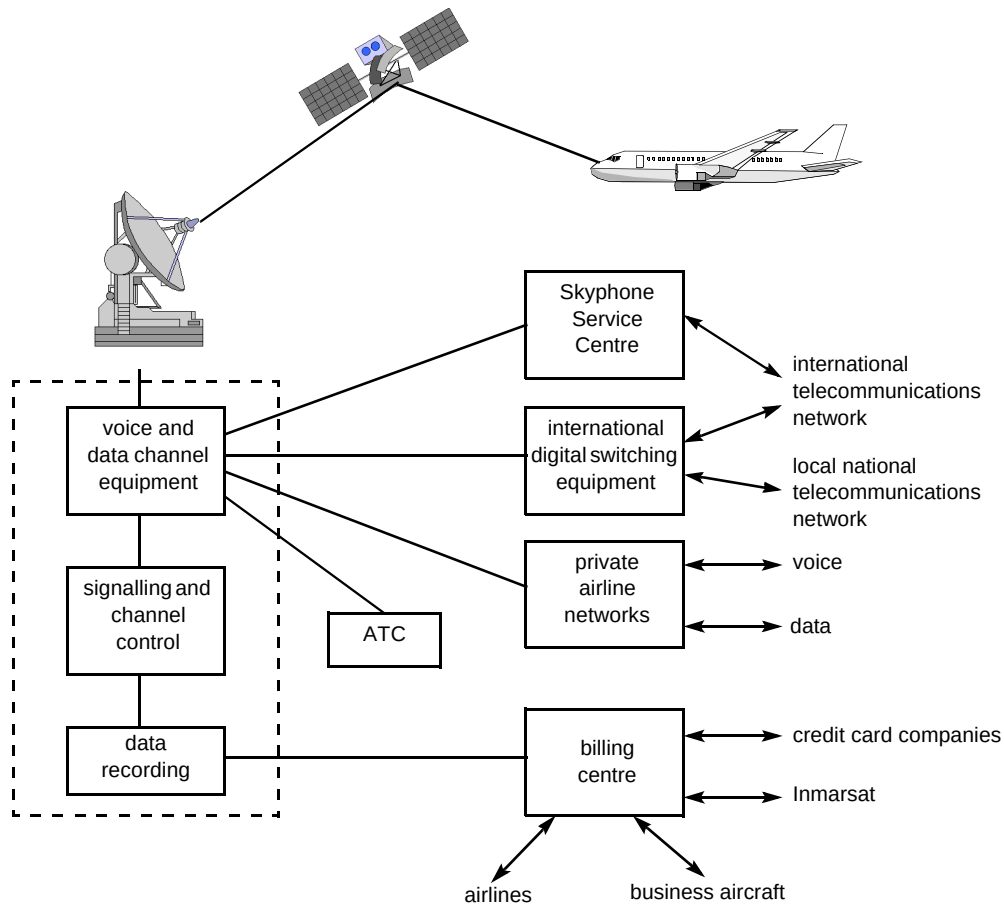


Fig 2 Satellite system architecture.

- R channels

The R channel is used for logging on to the system, requesting voice and data services and for sending short packet-mode data messages.

- T channels

The T channels are used primarily to carry packet-mode data. The receiving GES reserves time slots for transmissions requested by AESs, according to length. The sending AES transmits the message in the reserved time slots according to priority.

- C channels

The C channels are used in both forward and return directions to carry digital voice or data/facsimile traffic.

2. The Terrestrial Flight Telecommunications System

2.1 Introduction

The Terrestrial Flight Telecommunications System (TFTS) employs the digital cellular system concept and has been developed for the European market to offer low-cost telecommunications services to aircraft operating over land.

Table 1 The four RF channels between the aircraft and the ground stations.

Name	Access method	Modulation	Direction	Bit rates (bit/s)
P Channel	time division multiplex (TDM)	A-BPSK for low rates A-QPSK for high rate	ground-to-air	600, 1200, 10 500
R Channel	random access slotted Aloha	A-BPSK for low rates A-QPSK for high rates	air-to-ground	600, 1200, 10 500
T Channel	time division multiplex access (TDMA)	A-BPSK for low rates A-QPSK for high rate	air-to-ground	600, 1200, 10 500
C Channel	single carrier per channel (SCPC)	A-BPSK for low rates A-QPSK for high rate	bi-directional	21 000

A-BPSK (Aviation Binary Phase Shift Keying) is a form of differentially encoded BPSK and A-QPSK (Aviation Quaternary Phase Shift Keying) is a particular form of offset QPSK.

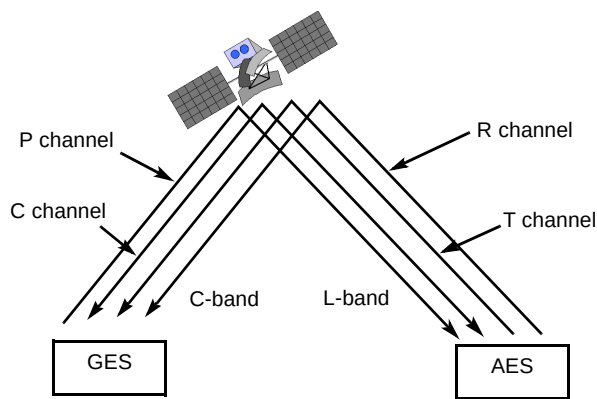


Fig 3 Inmarsat channel types.

It enjoys significant cost savings over satellite-based systems, offering similar services and features together with greater system capacity.

Commercial services based on this European Telecommunications Standards Institute (ETSI) standardised system have been operational since February 1995 when BT opened its network to its first customer — British Airways. Since this date, commercial services have opened in Sweden, Denmark, Finland and France. Many other European countries have plans in place for the roll-out of their TFTS networks.

2.2 Services and facilities

The TFTS was originally conceived to offer a basic air-to-ground telephony service on passenger aircraft; indeed the service is referred to in some circles as an Aeronautical Public Correspondence (APC) system. Since its conception, the system has grown in complexity — through the addition of further market requirements — to encompass not only the speech service, but paging and engineering paging features. Furthermore, the system has been designed to be enhanced to support circuit-mode data, packet-mode data and facsimile services. These latter features are the subject of ongoing standardisation effort and may enable innovative operators and/or service providers to offer ground-to-air calling capabilities should this become commercially viable.

To simplify integration with the satellite system, the telephony service is provided through the use of the same codec used in the Inmarsat system (see section 1.3).

The **paging** facility has been developed to enable communications to be established between ground systems and aircraft systems or aircraft passengers. Although commercial products have yet to be offered, this facility offers the potential for support of messaging systems and linkages to terrestrial systems offering electronic mail and messaging services.

The **engineering paging** facility has been developed as a part of the TFTS to enable aircraft equipment manufacturers to link directly to their equipment on-board aircraft for fault diagnosis, testing and similar purposes.

2.3 Principal system features

In common with land mobile cellular systems, TFTS has been defined from the start to support continuity of calls across networks through the handover capability, whereby the point of attachment to the network (the radio channel) is changed as the mobile terminal (the aircraft) moves through the network. However, where land mobile systems limit the handover feature to single networks (i.e. once a mobile terminal reaches the limit of the geographic coverage of the network the call ceases), TFTS has been designed to provide continuity of calls throughout the coverage area, regardless of the network providing the coverage.

TFTS is thus said to support **international handover**, since the general case of an internetwork handover will be between countries (e.g. London — Paris flights may require handover between BT and France Telecom TFTS networks).

Each aircraft has at least one transceiver supporting up to four simultaneous calls (see section 2.5). These calls can start or end completely independently.

2.4 System architecture

Figure 4 shows the architecture of the TFTS network. Readers familiar with the GSM system architecture will note the similarities between the two systems. Indeed, GSM standards were used as a basis for the standardisation of the TFTS system. However, a number of key differences can be demonstrated:

- TFTS supports multiple simultaneous calls between a mobile (aircraft) station (AS) and the ground network (i.e. the AS provides a multiplexing function),
- TFTS supports handover of multiple calls from a single aircraft,
- TFTS supports cell sizes of up to 400 km radius with a typical radius of 240 km,
- TFTS has been designed for aircraft speeds of up to 1100 km/h and associated Doppler shift,
- different ground station types have been specified for the different altitude scenarios covering aircraft altitudes up to 43 000 ft,
- TFTS does not support database (HLR, VLR) functionality separate from the switch (GSC) — all checks on access permission are made within the GSC,

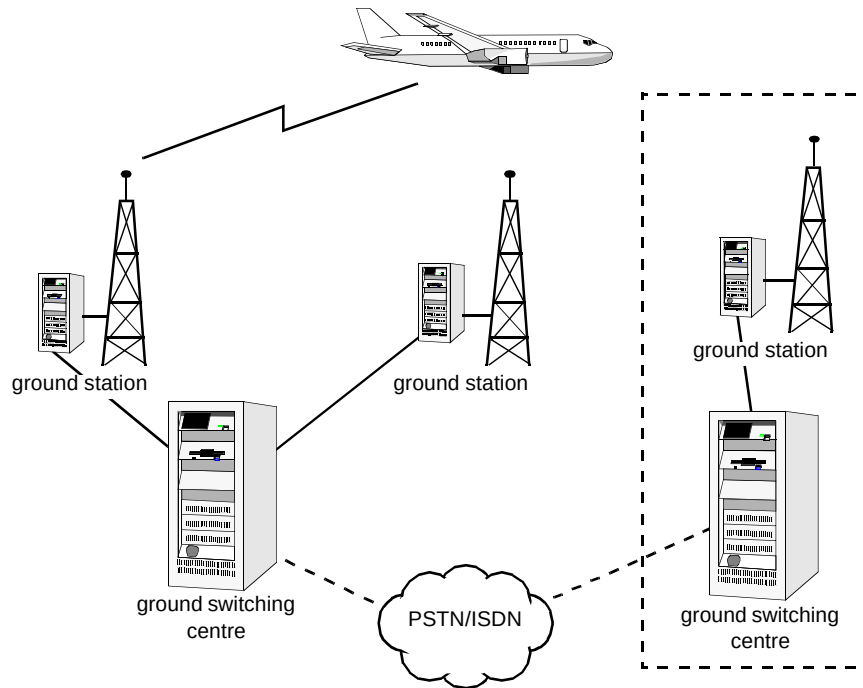


Fig 4 TFTS architecture.

- TFTS supports only air-to-ground calls — therefore location management functions are unnecessary,
- the GS and GSC can be combined into a single network node with the GS-GSC interface subsumed,
- TFTS supports international (internetwork) handover.

2.5 Radio interface structure

The TFTS radio interface is a full duplex TDM/TDMA and operates in the 1.6 to 1.8 GHz band with the following frequencies allocated on a world-wide basis:

1670 — 1675 MHz	ground-to-air use
1800 — 1805 MHz	air-to-ground use

This 2×5 MHz spectrum supports 164 channels, each of 1/33 MHz. Each channel supports a frame comprising 17 time-slots with a duration of 80 ms and a gross bit rate of

44.2 kbit/s. Each time-slot thus contains 208 bits and has a duration of approximately 4.706 ms. Figure 5 gives a schematic representation of the frame structure utilised in the TFTS.

The frames are organised into a 20-frame superframe which enables the support of user channels at gross bit rates of 2.4, 4.8 and 9.6 kbit/s. For the first phase of the TFTS, only channel rates at 9.6 kbit/s are supported, each comprising four time-slots. Thus the TFTS channel is capable of supporting four simultaneous duplex telephony channels, together with associated signalling channels.

Figure 6 shows the superframe structure supported by the TFTS radio interface. From Fig 6 it can be seen that timeslots 0 to 15 support traffic while timeslot 16 is variously used to support control channels, broadcast (synchronisation) channels (uplink part) and random access channels (downlink part).

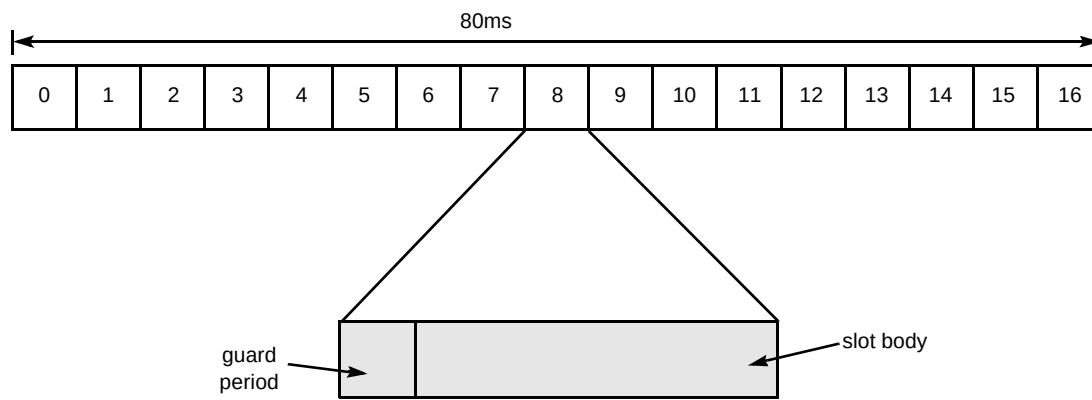


Fig 5 TFTS time-slot and frame structure.

0	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	B(S)/R
1	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
2	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
3	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
4	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
5	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	B(S)/R
6	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
7	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
8	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
9	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
10	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	B(S)/R
11	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
12	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
13	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
14	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
15	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	B(S)/R
16	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
17	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
18	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C
19	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	T	C

Fig 6 TSTS superframe structure.

The following channel types are supported.

- TCH (traffic channel) — these support the bi-directional carriage of user information (data) and can occupy time-slots 0 to 15, with each time-slot supporting a gross bit rate of 2.4 kbit/s (192 bits in 20 ms).
- DCCH (dedicated control channel) — this supports all signalling requirements between aircraft station (AS) and GSS and is made up of a combination of fast associated control channels (FACCH) and/or slow associated control channels (SACCH). A single DCCH is available between each AS and the ground network with an aggregate rate made up from the rates on each of the contributing channels (FACCH and SACCH).
- FACCH — these channels contribute to the aggregate data rate available on the DCCH and can occupy time-slots 0 to 15 when they are not being used for traffic (only immediately prior to connection of the traffic channel during the call set-up phase and immediately after disconnection of the traffic channel during the call clearing phase).
- SACCH — these channels contribute to the aggregate data rate available on the DCCH and can occupy time-slot 16 only on a timeshare basis during the superframe. Put simply, time-slot 16 is used for the SACCH associated with each time-slot in the range 0 to 15 once and only once during the superframe. Hence if all time-slots 0 to 15 are allocated to an aircraft simultaneously, the aggregate rate on the SACCH will be $16/20 \times 2.4 \text{ kbit/s} = 1.92 \text{ kbit/s}$.
- BCCH(D) — this channel type supports the broadcast of network data for use by the AS in selecting a network access point and basic radio parameters for the access. This channel type occupies one time-slot per GS which is selected from the time-slots 0 to 15 according to traffic usage. It should be noted that when a GS has allocated all its available time-slots for traffic, the BCCH(D) is not used; this indicates to all aircraft that the GS is full.
- BCCH(S) — this channel type supports the broadcast of synchronisation information from the network to the AS for the purposes of enabling initial access to the network. The BCCH(S) is transmitted in time-slot 16 of the frame in frames 0, 5, 10 and 15 of the superframe. The BCCH(S) carries information to enable aircraft stations to find the BCCH(D).
- RACH — this channel type supports initial access attempts by the aircraft stations into the ground network for the purposes of call establishment.

- IRCCH — this channel type supports the transmission of initial (radio) control information from the ground network to the aircraft stations following initial access using the RACH. This will include power and timing control information.
- RCCH — this channel type supports the ongoing transmission of control information from the ground network to the aircraft stations during ongoing connections for the purpose of power and timing control adjustments.
- ICH — this channel type is transmitted by the GS in all unused time-slots (i.e. where no other channel type is using the time-slot).

2.6 Network aspects

The TFTS system has been defined [4—6] as a network of ground station systems (GSS), each comprising one or more ground stations (GS) connected to a single ground switching centre (GSC). Each GS serves a limited geographical area termed a cell. The combination of multiple cells provides wide area geographic coverage through the reuse of radio frequencies, according to the accepted principles of cellular radio communications.

The GSC constitutes the interface between the TFTS radio sub-system and the public telecommunications networks. It performs all the switching functions for ASs located in a geographical area designated as the GSC area. The main difference between a GSC and an exchange in the fixed network is that the GSC takes into account the impact of the allocation of radio resources and the mobile nature of subscribers. In addition, it performs at least the following procedures:

- call set-up,
- aircraft access validity check,
- handover between radio channels within a GSC,
- handover between radio channels of different GSCs,
- operations and maintenance (O&M),
- assimilation of data for billing of network usage.

The GS is a network component which comprises the transceiver equipment needed to establish the radio link between the ground network and the AS. The GS serves one cell and is controlled by the GSC.

There is also an interworking function (IWF) associated with the GSC. The IWF provides the necessary functionality to allow interworking between the TFTS network and the fixed networks (i.e. PSTN, ISDN and

PSPDN), connecting between the protocols used in the TFTS network and those used in the fixed networks. The IWF may have no functionality where the service implementation in the TFTS is directly compatible with that in the fixed network.

3. Future aeronautical services

3.1 Skyphone

The next major development for the Inmarsat aeronautical system will be the introduction of a new reduced rate voice codec. Inmarsat have selected Digital Voice Systems Inc's (DVSI) advanced multi-band excitation (AMBE) codec which operates at an aggregate rate of 4.8 kbit/s.

The new codec will initially be used with the Aero-H system and will allow a single aircraft installation to support up to ten voice channels. In the future, the codec will form an integral part of the new Aero-I system that is currently under development.

Aero-I will use an intermediate gain antenna of 6 dBi and a reduced-rate C-channel operating at 8.4 kbit/s; this compares with the existing voice channel operating at 21 kbit/s. The attraction of Aero-I is that it will require less bandwidth, down from 17.5 kHz to 5 kHz, resulting in lower satellite charges. Furthermore, the cost and the weight of the avionics equipment should be significantly lower with the use of a smaller and less sophisticated antenna.

However, Aero-I has been designed specifically to operate using the spot beam technology available on Inmarsat's next generation of satellites. During 1996, the first Inmarsat-3 satellite will be launched and will provide up to five spot beams. Within a spot beam, Aero-I will offer 4.8 kbit/s telephony, 2.4 kbit/s telefax and 4.8 kbit/s data services. In the global beam only a basic 600/1200 bit/s data service will be available.

Aero-I is intended for the short-haul aircraft and thus is a potential competitor to TFTS and other regional systems.

In contrast to the Aero-I system, Inmarsat are developing a broadcast high-speed data service for possible aeronautical use. The system is very flexible, supporting throughputs from 2 kbit/s up to 70 kbit/s. Furthermore, the capacity can be dynamically shared between different services, e.g. teletext, slow-scan video, and radio.

3.2 TFTS

As discussed above TFTS currently supports little more than a telephony service. However, moves are afoot within ETSI to develop further services. In common with other aspects of telecommunications, customers are increasingly expecting to be able to communicate through means other

than speech while on the move. Therefore, key among the candidate services are circuit-mode data and facsimile services.

The aeronautical community, in common with other transportation industries, is not just in the business of moving people from one point to another — increasingly they are looking to generate revenue from their temporarily captive audience through the use of a variety of value-added services. Furthermore, the major players are keen to offer their passengers more than just a transportation service in order to maintain and enhance their competitive positions.

BT has played a major role in enabling airlines to meet these more complex needs through the provision of the ALIS range of value-added services, currently provided to the satellite aeronautical community through the use of packet mode communications. TFTS has the capability to offer similar functionality to short- and medium-haul aircraft.

Facsimile and circuit-mode data

Since the early 1980s facsimile has grown dramatically to the point where it is almost indispensable as a business tool. It is the expectation of the aeronautical community that once the use of telephones on aircraft becomes commonplace and accepted business practice, the desire for facsimile will follow.

In a similar manner, the explosive growth of data communications (driven for example by the use of e-mail) is expected to be mirrored on aircraft, though perhaps to a lesser degree.

ETSI is therefore seeking to provide cost-effective solutions for the provision of facsimile and circuit-mode data to the aeronautical community through the TFTS. Although more than one candidate solution is being considered, the enhancements made to the speech codec already in use in TFTS for the Skyphone system are being given serious consideration; however, because of fundamental differences in the system concepts there are specific issues to be overcome if this codec-based solution is to be adopted.

Provided these problems can be resolved, it will be possible for airlines to offer facsimile and circuit-mode data communications to their passengers before mid-1997.

Packet-mode data

Since 1994 ETSI has been working to develop a supplement to the TFTS standard to support the transport of packet-mode data between suitably equipped aircraft and the ground network. The standard will further support the

flexible interworking with a variety of data networks including public and private data networks.

Currently, the appropriate sub-technical committee of ETSI (STC RES5) has approved the draft standard for the provision of packet data. Subject to the ratification of this decision by the Technical Committee, the draft standard will begin public enquiry in mid-1996, and will be published in the first half of 1997, provided the public enquiry is successfully negotiated.

It is envisaged that the service providers who sell the capabilities of the TFTS to airlines and their customers will drive the innovative services which will be delivered using the TFTS packet-data capabilities.

4. Current commercial uses for aeronautical systems

4.1 Skyphone

More than 700 aircraft have already been fitted with Inmarsat aero terminals and from these almost 200 corporate aircraft and 350 airliners are commissioned to use the system. Some 40 international carriers now operate aircraft equipped for satellite communications and offer passengers phone and facsimile services. In the next 18 months a further 1200 aircraft will be fitted out, enabling up to 80 % of passengers on long-haul routes to benefit from Inmarsat aeronautical mobile satellite communications.

4.2 TFTS

The market structure for the sale of TFTS services to the airline community has been dictated to some degree by the EU directive on competition in telecommunications. As a result, TFTS network operators are prevented from selling direct to the market (airlines) and must do so through service providers.

At present, the service provider Jetphone (50/50 joint venture between BT and France Cables et Radio — a division of France Telecom) has two major customers with the following number of aircraft equipped:

- British Airways — 11 aircraft operating on UK routes (BA were the world's first commercial user of TFTS),
- Scandinavian Air Services (SAS) — committed to equipping their complete European fleet of 105 aircraft for use on European routes.

Currently these provide telephony service only.

It is understood that a number of other airlines are considering offering their passengers the facilities of TFTS and are in varying stages of negotiating contracts for the provision of service.

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working on administration, numbering and billing aspects of GSM.

Jim Pettifor graduated from the University of Bath in 1987 with a degree in Physics with Physical Electronics. Following a brief period in materials research, he joined BT Laboratories in 1989 to work on mobile communications systems. He has worked on a variety of network aspects of cellular systems and since 1990 has played a key role in the development of the TFTS, including acceptance testing of TFTS network components leading to the launch of BT's TFTS network. Since 1994, he has led BT's involvement in the development of packet-mode services on TFTS. He is currently



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